UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 09 — Optional

Root locus, gain, uncertainty and actuator limits

Optional self-study after Lecture 8. This tutorial is outside the eight-class core route and is separate from Project 2.
Optional independent practice: start with Q1, then Q2–Q3. Sections labelled Extension or After class are independent practice. Show reasoning, substitutions and units.

Question 1 Quick checks on root-locus meaning and feasible commands

Give a brief reason or calculation.
a) Root-locus endpoints. For unity negative feedback with $L(s)=K(s+1)/[(s+2)(s+4)]$, K ≥ 0, where do the two branches start and where do they end as K tends to infinity?

b) Close a simple loop. For $L(s)=K/(s+1)$, find the closed-loop pole at K = 1 and K = 4. What happens to this pole as K increases?

c) Read a pole pair. A standard constant-numerator second-order model has poles $-3\pm j4$ s^{-1}. Find natural frequency, damping ratio and the approximate 2% settling time.

d) Actuator clipping. A stage obeys $\ddot{x}+3\dot{x}=u$, with $u=25(r-x)-3\dot{x}$. From rest apply r = 0.020 m, but cap positive u at 0.3 m/s². Find requested command, delivered command and actual initial acceleration.

Question 2 Design a velocity-feedback inspection stage

A calibrated inspection stage satisfies $x'' + 3x' = u$, where x is in metres and u is an acceleration-equivalent command in m/s². It is not a force or motor voltage. Compare P control $u = K_p(r - x)$ with velocity feedback $u = K_p(r - x) - K_v x'$. The target poles are $-3 \pm j4\ s^{-1}$. The reference is a 0.020 m step from rest. Requirements are overshoot ≤10%, approximate 2% settling time ≤1.5 s, and initial command ≤0.6 m/s².

a) Match the target polynomial and find K_p and K_v, including their units.

b) Compare poles, damping, overshoot, peak time and estimated settling time with P-only control using the same K_p.

c) Check the nominal specifications and initial command. Which claim would require checking the full command trajectory?
MATLAB verification. After matching the poles, compare $K_p = 25$ with $K_v = 0$ and 3 for the 0.020 m step from rest. Submit one figure with position and requested command in separate panels; distinguish both controllers and mark 0.6 m/s². Report poles, overshoot, settling and initial command. The simulations are unsaturated. Base MATLAB is sufficient.

Question 3 A nominally good design with uncertain damping

Keep $K_p = 25$ from Question 2, but the actual physical velocity coefficient b in $x'' + bx' = u$ can lie anywhere from 2.4 to 3.6 s^{-1}. First use $K_v = 3.0\ s^{-1}$. Then redesign only K_v to preserve overshoot ≤10% and approximate 2% settling time ≤1.5 s across this range.

a) Find the worst damping ratio and the corresponding overshoot and settling-time estimates for K_v = 3.0. Does the nominal design remain acceptable?

b) Translate both requirements into lower bounds on K_v. Choose a simple value meeting both and verify the worst case.

c) Recheck initial command for a 0.020 m step. State what the uncertainty calculation has and has not established.

Extension A Higher gain does not always move the dominant pole left

After class
The separate normalized plant below is under unity negative feedback with proportional gain K > 0. Compare K = 1 and K = 4.

$$G(s) = \frac{s+1}{(s+2)(s+4)}$$

a) Derive the closed-loop characteristic polynomial and calculate both pole pairs.

b) Identify the dominant pole and its trend as K increases. Explain the role of the plant zero without needing detailed root-locus construction rules.
c) Calculate unit-step steady errors and reconcile better steady accuracy with the pole trend.

Method reference

Quick-check sources: Ogata, Modern Control Engineering, 5th ed., Sec. 6–2, pp. 270–278, and Sec. 5–3, pp. 165–177. Q1 is a new concept-level adaptation. The previous actuator check is retained as Q1(d).
Method source: Ogata, Modern Control Engineering, 5th ed., Problem A-5-4, pp. 235–236 (velocity feedback); Chapter 6 (root-locus interpretation). The uncertainty test and gain family are independently adapted. Page numbers refer to printed pages.
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